Young's Systems Development Consulting Inc.

Memo

To: Tim Smith

From: Mallory Young

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Re: A Summary and Comparison of DevOps and SDLC

What is DevOps?

DevOps is short for Development Operations, and it is a software engineering culture that unifies the development team and the operations team. It includes best practices and tools that support continuous integration and deployment. DevOps uses an integrated set of tools (mostly open source) to automate software delivery. The success of DevOps relies on the development team and the operations team working together. As part of this process, the development team will submit the application to the operations team for implementation. DevOps is broken down into an eight-phase cycle. These eight phases can be broken down into 2 categories. First, the development stage includes the Plan, Code, Build, and test phases of DevOps. The operations stage includes the Deploy, Operate, and Monitor phases. The 8 phases of DevOps are briefly described below:

- *Plan*: The client and the software development team discuss the goals and create a plan for the development of the application.
- *Code*: Programmers design and code the application using tools that allow multiple programmers to work on the same code simultaneously.
  - Both the Plan and Code phases provide a repository for managing and storing various versions of a code.
- *Build*: Programmers take the sections of code from different repositories and combine them to build a complete application.t
- *Test*: The application is tested using automated testing tools to ensure software quality. Testing is automated within the build pipeline to ensure the application is deployable.
- *Integrate*: When testing is complete, the Integrate phase takes new features and integrates them automatically into the existing code base.
- *Deploy*: The application is packaged after its release and deployed from the development server to the production server.
- *Create*: Involves the operations team performing activities such as configuring servers providing the required resources for the application to run smoothly.

- *Monitoring*: Includes continuously monitoring the application after release for spikes and issues that are relevant to the code or specific consumer habits that the client wishes to monitor.

DevOps and the SDLC

DevOps is not used as a replacement for an SDLC. They are different concepts and do not compete with each other. They do however complement each other, and DevOps can be embedded within the SDLC for optimization of the cycle. One of the advantages of incorporating DevOps into the SDLC is that it accelerates the project life cycle which gets the product to the client sooner and will in turn increase profits. DevOps works with the SDLC by integrating the communication of the development team and operations teams while also integrating automation tools in as many phases of the cycle as possible.

From my research I have found that the DevOps phases augment or sometimes replace the SDLC phases in the following ways:

The Requirements/Analysis and the Design phases of the SDLC both align with the Plan phase of DevOps. In the SDLC Requirements/Analysis phase developers meet with the client to obtain the relevant information that is used to develop an application specific to their expectations. In the SDLC Design phase, analysts convert the information gained from the client during the Requirements/Analysis phase to create logical and physical systems specifications. Both of these phases are augmented with the DevOps Plan phase when the client and the software development team meet to discuss the goals and create a plan for the development of the application.

The Development phase of SDLC is sometimes referred to as Implementation. This phase aligns with the Code and Build phases of the DevOps cycle. The Development phase takes the logical and physical system specifications and translates them into source code to create a working system. During the Code phase of DevOps programmers design and code the application using tools that allow multiple programmers to work on the same code simultaneously. In the Build phase of DevOps programmers take the sections of code from different repositories and combine them to build a complete application.

The Testing/Quality Assurance phase of the SDLC aligns with the Test and Integrate phases of DevOps. Testing for SDLC involves testing the internal code and the functionality of the application as a whole for issues, bugs, vulnerabilities, etc. Quality Assurance is determined when evaluating the verification and validation of the application. During the Test phase of DevOps, the application is tested using automated testing tools to ensure software quality. When testing is complete, the Integrate phase takes new features and integrates them automatically into the existing code base.

Lastly, the Deployment phase of the SDLC aligns with the Deploy, Create, and Monitoring phases of DevOps. The Deployment phase of the SDLC involves releasing the application to the client and conducting user acceptance testing. Once final approval for the application is obtained, then maintenance of the application is continued to address any issues that may arise. In the Deploy phase of DevOps, the application is packaged after its release and deployed from the development server to the production server. The Create Phase involves the operations team performing activities such as configuring servers providing the required resources for the application to run smoothly. Finally, the Monitoring phase of DevOps includes monitoring the application after release for spikes and issues that are relevant to the code or specific consumer habits that the client wishes to monitor.

The chart below shows how each of the phases of the SDLC and DevOps are aligned/augmented:

SDLC	DevOps
Requirements/Analysis	Plan
Design	
Development	Code
	Build
Testing/QA	Test
	Integrate
Deployment	Deploy
	Create
	Monitoring

References

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